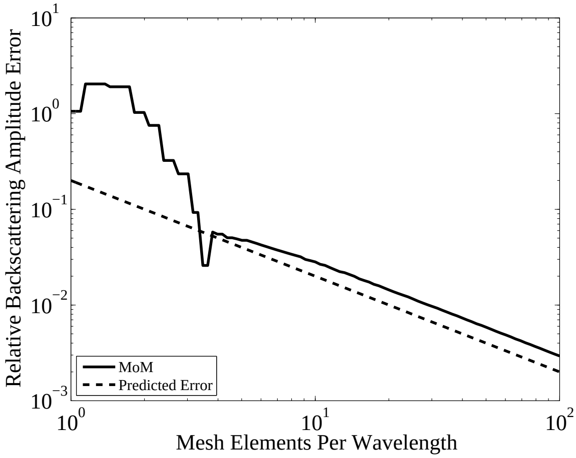


*Figure 6.7 Relative backscattering amplitude error for MoM applied to the TM-EFIE for a PEC circular cylinder with half wavelength radius. The MoM implementation uses point matching and a single-point quadrature rule for the source integration. The predicted error is computed using* (6.105). *The scattering amplitude computed with this MoM implementation converges at a first-order rate with respect to the number of mesh elements. The inaccurate single-point integration rule obscures the variational property of the method of moments. With a better integration rule for moment matrix elements, the accuracy of computed scattering amplitudes improves dramatically to third order.*

For homogenous dielectric objects, either surface or volume integral equations can be used. The surface integral formulation for a homogeneous dielectric consists of two integral operators, one similar to the operator in the EFIE and the other to the MFIE operator, and both electric and magnetic equivalent surface currents are required. For inhomogeneous objects, volume integral equations must be used, and the unknown current is nonzero on the region of support of the scatterer. Since the surface integral operators for homogeneous dielectrics are fairly similar to the integral equations for PEC objects we treated in the previous section, we will focus here on the volume integral formulation.

### *6.6.1 2-D Volume Method of Moments*

To derive the volume EFIE, we want to rearrange Maxwell's equations so that the permittivity and permeability are equal to their free-space values and the dielectric properties of the scatterer are lumped into an equivalent source. We will assume that the scatterer is nonmagnetic, so that $\mu = \mu_0$. By rearranging (4.114b) slightly, Faraday's and Ampère's laws can be written in the form

$$\nabla \times \overline{E}^{\mathrm{s}} = -j\omega\mu_0\overline{H}^{\mathrm{s}} \tag{6.106a}$$

$$\nabla \times \overline{H}^{\mathrm{s}} = j\omega\varepsilon_0\overline{E}^{\mathrm{s}} + \overline{J} \tag{6.106b}$$

where

$$\overline{J}(\overline{r}) = \{j\omega[\varepsilon(\overline{r}) - \varepsilon_0] + \sigma(\overline{r})\}\overline{E} \tag{6.107}$$

and $\overline{E}$ is the total field. With the governing equation (4.114b) used for the FDTD in the scattered field formulation, we had a known source radiating in the presence of the scatterer, whereas in (6.106) we have an unknown source radiating in the absence of the scatterer. This highlights a key difference between the FDTD algorithm, for which sources are known, and MoM, for which sources are unknown.

Because we have moved the effect of the scatterer into the equivalent current source, so that the source lies in free space, we can use the free-space radiation integral to express the scattered field in terms of the source. This leads to an integral equation for the unknown source. For a 2-D problem and $\mathrm{TM}^z$ polarized incident field, the scattered field radiated by the source is

$$E_z^{\mathrm{s}}(\overline{\rho}) = -\frac{k\eta}{4}\int_S d\overline{\rho}'\, H_0^{(2)}(k|\overline{\rho} - \overline{\rho}'|)J_z(\overline{\rho}') \tag{6.108}$$

Using the scattered field decomposition in (4.103), this can be rewritten to include the known incident field, so that

$$E^{\mathrm{i}}(\overline{\rho}) = E_z(\overline{\rho}) + \frac{k\eta}{4}\int_S d\overline{\rho}'\, H_0^{(2)}(k|\overline{\rho} - \overline{\rho}'|)J_z(\overline{\rho}') \tag{6.109}$$

or, in terms of the unknown current,

$$E^{\mathrm{i}}(\overline{\rho}) = \frac{J_z(\overline{\rho})}{j\omega\varepsilon_0[\varepsilon_r(\overline{\rho}) - 1] + \sigma(\overline{\rho})} + \frac{k\eta}{4}\int_S d\overline{\rho}'\, H_0^{(2)}(k|\overline{\rho} - \overline{\rho}'|)J_z(\overline{\rho}') \tag{6.110}$$

This integral equation has the form of a second-kind integral equation, but is not a true second-kind equation in the Fredholm classification because the kernel is not square-integrable. Since $\varepsilon(\overline{r}) = \varepsilon_0$ and $\sigma(\overline{r}) = 0$ outside the scatterer, $\overline{J}$ is nonzero only inside the scatterer, so the domain of the integral equation is the volumetric region occupied by the scatterer.

This integral equation can be solved by dividing the scatterer into $N$ small cells as shown in Figure 6.8 and expanding $J_z$ in terms of the 2-D pulse function

$$f_n(\overline{\rho}) = \begin{cases} 1 & \text{if } \overline{\rho} \text{ in cell } n \\ 0 & \text{otherwise} \end{cases} \tag{6.111}$$

We will test the $\overline{\rho}$ dependence with delta functions at the centers of each cell. The $\overline{\rho}'$ integrals can be evaluated using a single integration point for different testing and expansion cells, and the self-terms can be integrated analytically by approximating the cells as circular. If this is done, a linear system is obtained with matrix elements and the right-hand side given by

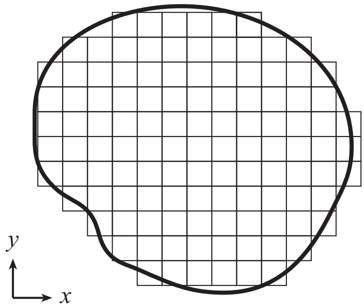


*Figure 6.8 2-D inhomogeneous dielectric scatterer with grid cells for the 2-D volume method of moments.*

**2-D Volume MoM for TM$^z$ Polarized Fields**

$$b_m = E_z^{\mathrm{i}}(\overline{\rho}_m) \tag{6.112}$$

$$A_{mn} = \begin{cases} \frac{\eta\pi s_n}{2} J_1(ks_n) H_0^{(2)}(k|\overline{\rho}_m - \overline{\rho}_n|) & m \neq n \\ \frac{\eta\pi s_n}{2} H_1^{(2)}(ks_n) - \frac{j\eta\varepsilon_{rn}}{k(\varepsilon_{rn}-1)} & m = n \end{cases} \tag{6.113}$$

where $s_n$ is the radius of a circle with the same area as the $n$th mesh element. To account for the conductivity of a lossy scatterer, the relative permittivity in this expression is taken to be the complex permittivity

$$\varepsilon_{r,c} = \varepsilon_r + \frac{\sigma}{j\omega\varepsilon_0} \tag{6.114}$$

By using (6.107), the total field on the scatterer can be found directly from $J_z$. If the scattered far field is desired, that can be found in postprocessing using (6.108). Using the same approximations as in (6.113), the scattering width can be written as

$$\sigma(\phi^{\mathrm{s}}) = \frac{k\eta^2}{4} \left| \sum_{n=1}^{N} a_n \frac{2\pi s_n}{k} J_1(ks_n) e^{j\overline{k}^{\mathrm{s}} \cdot \overline{\rho}_n} \right|^2 \tag{6.115}$$

where $a_n$ are the current unknowns resulting from the solution of the linear system with the right-hand side having elements (6.112) and matrix given by (6.113).

Using similar methods, we can derive the 2-D volume integral equation for the TE polarization and the 3-D volume electric field integral equation. Unlike the 2-D TM$^z$ polarization that we have considered above, these integral equations involve derivative operators, which increases the singularity of the integrand, and specialized integration techniques are needed to implement the method of moments. For the 3-D volume integral equation, the mesh is three-dimensional and the mesh elements are generally cubes or tetrahedra.

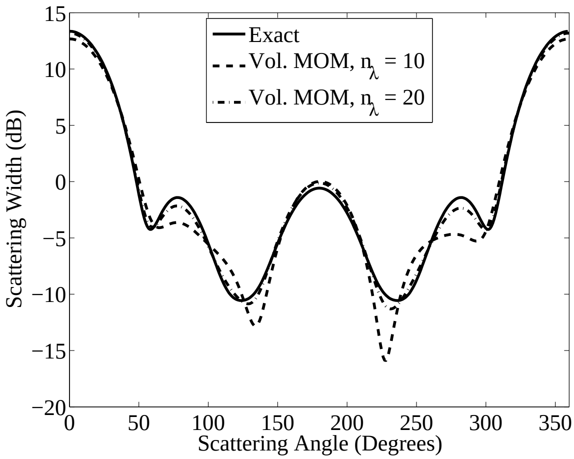


*Figure 6.9* *Bistatic scattering width of a dielectric cylinder with radius $a = 0.5\,\lambda$ and relative permittivity $\varepsilon_r = 2$. The analytical solution is compared with the numerical solution for two different mesh discretization densities.*

6.6.1.1 Numerical Example

Computed and analytical bistatic scattering width values are shown in Figure 6.9 for a dielectric cylinder illuminated by a $TM^z$ polarized plane wave arriving from the incidence angle 180°. Numerical results are given for two values of the mesh discretization density. For $n_\lambda = 10$, the mesh has 73 elements, and for $n_\lambda = 20$, the mesh has 309 elements. Accuracy is not quite as good as the surface method of moments with the same mesh element density, because the volume MoM mesh is made up of square elements and the effect of geometrical modeling error is more significant than is the case for the surface MoM.

## 6.7 2.5-Dimensional Methods

All the numerical methods we have looked at so far are two-dimensional, because the fields and sources are constant in the $z$ direction. Three-dimensional algorithms require vector basis functions, because the fields and currents cannot be treated as scalar quantities. Meshing a three-dimensional object can also be challenging. Before considering three-dimensional problems, it is worth observing that there is an intermediate class of problems for which the structures (conductors and dielectrics) are two-dimensional but the fields are three-dimensional. This means that the illuminating field strikes an infinitely long, cylindrical object obliquely with respect to the axis of symmetry of the object. Two-dimensional codes can be modifies to accommodate the $z$ variation of the incident field. These methods are sometimes referred to as 2.5-dimensional (2.5-D).